

Performance and Security Evaluation of Federated Learning under IID and Non-IID Settings and Differential Privacy

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Abstract—Federated Learning (FL) has emerged as a promising approach for privacy-preserving machine learning by enabling multiple clients to train a global model collaboratively without sharing their raw data. Despite its decentralized nature, FL remains vulnerable to inference attacks due to the exposure of model updates. In this article, we comprehensively evaluate federated learning in IID and non-IID settings and consider the exchange between model performance and data distribution. Besides, we proposed a privacy-enhanced FL framework that integrates Gaussian-mechanism-based Differential Privacy (DP) at the client side, by injecting calibrated noise into local model parameters before aggregation. Our method ensures a tunable trade-off between utility and privacy while preserving the original FL architecture. Experiments on the CIFAR-10 dataset using the Vision Transformer (ViT-Tiny) show that while Centralized Learning (CL) achieves the highest accuracy, FL with DP can effectively preserve privacy with moderate performance overhead. These findings offer practical guidance for deploying federated learning systems that balance performance and privacy in real-world scenarios.

Index Terms—Federated Learning, Differential Privacy, Non-IID, Gaussian Noise, Privacy-Preserving, Vision Transformer.

I. INTRODUCTION

The proliferation of smart devices such as smartphones, sensors, and cameras has accelerated the adoption of decentralized machine learning at the network edge. These devices continuously generate large volumes of heterogeneous data that fuel intelligent services. However, uploading raw data to centralized servers for training introduces substantial communication overhead, latency, and privacy risks—especially when sensitive user data is involved [1].

Federated Learning (FL) has emerged as a promising paradigm to address these limitations by enabling clients to collaboratively train models without sharing their raw data [2]. While this design mitigates

privacy exposure, model updates remain vulnerable to inference attacks, such as gradient inversion and membership inference. Moreover, FL often suffers from performance degradation in the presence of non-independent and identically distributed (non-IID) data, which is common in real-world deployments [3].

To enhance privacy, Differential Privacy (DP) mechanisms have been integrated into FL systems by injecting calibrated noise into model updates [4]. However, this introduces a fundamental trade-off: higher noise levels improve privacy but degrade accuracy. The challenge is exacerbated under data heterogeneity and sparse client participation, where tuning noise scale becomes nontrivial.

In this study, we introduce a modular FL framework that applies local DP through Gaussian noise addition to client-side model parameters before aggregation. Our system, built on the Flower framework and using ViT-Tiny as the backbone, supports flexible data partitioning schemes—including IID and label-based splits—and allows configurable noise scales for privacy control. On this framework, we employ several scenarios to evaluate FL in different contexts of data distribution and security.

Beyond these contributions, we explicitly structure our study around the following research questions (RQs):

- **RQ1 – Impact of decentralization:** How does federated learning with a ViT-Tiny backbone compare to centralized learning in terms of classification performance and training time on CIFAR-10?
- **RQ2 – Impact of data heterogeneity:** Under the same model and training budget, how do different data partitioning strategies (IID versus 80/10/10 label-skewed Non-IID) affect the convergence behavior and final performance of the

global federated model?

- **RQ3 – Privacy–utility trade-off under DP:** When client-side Gaussian noise is injected into local model parameters, how does the noise scale influence the trade-off between privacy and utility in both IID and Non-IID federated settings?

The main contributions of this paper are as follows:

- We propose a lightweight, privacy-aware Federated Learning framework that incorporates configurable, client-side Gaussian-mechanism-based DP, built upon the Flower framework and the ViT-Tiny architecture;
- We support realistic FL scenarios by implementing various data partitioning strategies;
- We perform a comprehensive evaluation along three key dimensions: decentralization, data heterogeneity, and privacy.

The rest of the paper is organized as follows: Section II reviews previous works on FL and related issues on data distributions and security; Section III outlines background concepts of Federated Learning, Vision Transformer Backbone, and Differential Privacy. Our proposal and methods are presented at Section IV; Next, Section V describes the experimental setup and results. The paper ends with our conclusion and future works.

II. RELATED WORKS

Federated Learning has emerged as a distributed learning paradigm enabling decentralized clients to collaboratively train models without sharing raw data. Its standard architecture follows a synchronous client-server model, where the server distributes a global model and aggregates local updates from selected clients. The foundational FedAvg algorithm [5] uses weighted averaging to aggregate client updates, offering communication efficiency but suffering from degraded performance under non-IID data due to client drift and optimization inconsistency.

To address this, several optimization strategies have been developed. FedProx [6] adds a proximal term to stabilize local updates. SCAFFOLD [7] introduces control variates to reduce variance, while MOON [8] employs contrastive learning to align local and global representations. FedNova [9] adjusts for variable local training durations, improving fairness and convergence.

In parallel, robust aggregation and client selection methods have been explored. Byzantine-resilient techniques such as Krum [10] and Median aggregation reduce the impact of malicious clients. Client selection schemes like FedCS [11] and Oort [12] prioritize participants based on availability, data quality, and system constraints.

For privacy preservation, DP is widely adopted in FL. DP-FedAvg [13] applies Gaussian or Laplace noise to ensure (ϵ, δ) -DP guarantees. Alternative frameworks such as PATE [14] aggregate teacher predictions instead of gradients, enhancing privacy at the cost of increased communication. Nonetheless, most DP-integrated FL approaches face an inherent trade-off between privacy and utility, particularly under non-IID conditions.

Meanwhile, ViT [15] has demonstrated strong global representation capabilities in vision tasks. Unlike CNNs, ViT processes images as sequences of patches via transformer encoders. While ViT excels in centralized settings, its adoption in privacy-sensitive and resource-constrained FL environments remains limited and underexplored.

Most prior work isolates either data heterogeneity or privacy preservation, while often neglecting system-level factors such as training time, resource usage, and model scalability—key considerations in real-world FL deployment.

In contrast, our work introduces a unified experimental framework that evaluates both performance and privacy under heterogeneous data conditions. Unlike prior DP-FedAvg or PATE-based approaches, we implement a lightweight local Gaussian-mechanism-based DP module and integrate it into a Vision Transformer backbone, prioritizing low overhead on federated clients. This design enables tunable privacy-utility trade-offs and practical development viability under constrained environments.

III. PRELIMINARIES

A. Federated Learning Framework

FL is a decentralized machine learning paradigm that enables collaborative training across distributed clients while preserving data locality [5]. In each communication round, the server sends the global model to K clients, each of which trains locally on its private dataset $\mathcal{D}_k$ and returns updated parameters. The server aggregates these using the Federated Averaging (FedAvg) rule in Eq. (1), following the standard FL formulation in [5], where $N = \sum_{k=1}^K |\mathcal{D}_k|$ denotes the total number of samples across all clients. This process is repeated for R rounds, progressively refining the global model while ensuring that raw data never leaves local devices.

$$\theta^r = \sum_{k=1}^K \frac{|\mathcal{D}_k|}{N} \theta_k^r \quad (1)$$

B. Vision Transformer Backbone

ViT [15] reformulates image classification as a sequence modeling task. Each input image is first resized to 224×224 and divided into fixed-size patches of 16×16 . These patches are flattened and

linearly projected into 768-dimensional embeddings. A special classification token ([CLS]) is prepended to the sequence, and positional encodings are added to preserve spatial information.

The resulting sequence is passed through multiple Transformer Encoder layers, each composed of Multi-Head Self-Attention (MHSA) and Feed-Forward Network (MLP) blocks with residual connections. The final [CLS] token output is passed to a dense layer with softmax for classification.

While ViT demonstrates superior generalization capabilities compared to CNNs—especially on large-scale datasets like ImageNet-21k—it demands substantial computational resources. Therefore, we adopt the lightweight ViT-Tiny variant as our backbone model in the FL system to balance performance and efficiency under constrained resources.

C. Differential Privacy via Gaussian Noise

In FL, model updates can still leak private information, such as membership inference or reconstruction of training data [16]. To mitigate this risk, Differential Privacy (DP) can be enforced by perturbing updates before aggregation. To mitigate these risks, DP introduces controlled randomization to limit the influence of any single data sample on the shared parameters.

Typical DP-FL methods perform ℓ_2 -norm clipping before applying the Gaussian mechanism to ensure a formal (ϵ, δ) -DP guarantee [4]. Inspired by these approaches, and to reduce client-side overhead, our framework adopts a lightweight, Gaussian-mechanism-based variant that directly perturbs local model parameters as in Eq. (2):

$$\tilde{\theta}_i = \theta_i + \mathcal{N}(0, \sigma^2 I) \quad (2)$$

where σ is the standard deviation of the Gaussian noise, and I is the identity matrix, indicating that independent noise is added to each parameter of θ_i . The noise level is configurable via the `noise_scale` parameter. Although omitting clipping means our method does not yield strict (ϵ, δ) -DP, it follows recent practical FL studies [17] emphasizing efficiency and convergence under heterogeneous data while reducing privacy leakage.

IV. PROPOSAL APPROACH

A. System Overview

We design a synchronous client-server FL system based on the Flower framework, supporting both standard and privacy-preserving configurations, and integrating flexible data partitioning strategies (e.g., IID, label-based). Each training session consists of multiple communication rounds. In each round, the server distributes the current global model to all participating clients. Clients train locally on private

subsets of the datasets using deep learning models and return their updated parameters. These updates are aggregated using the FedAvg strategy as defined in Eq. (1) to form a new global model for the next round.

To protect user privacy, our framework optionally adds Gaussian noise to the model parameters before transmission to the server. This DP mechanism is tunable via a configurable noise scale and can be selectively enabled per experiment.

Figure 1 shows the FL system architecture, highlighting the flow of model parameters and optional noise injection. Raw data never leaves the client device, ensuring inherent privacy-by-design principles throughout the learning process.

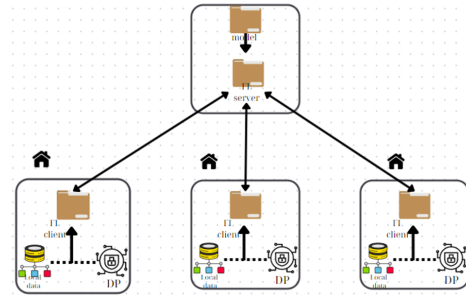


Fig. 1: System Architecture of Privacy-Aware Federated Learning Framework

Figure 2 illustrates the internal structure, which includes:

- **Preprocessing:** Images are resized to 224×224 , augmented, and normalized.
- **Patch Embedding:** Each image is split into 196 patches of size 16×16 , flattened into 768-dimensional vectors.
- **[CLS] Token:** A classification token is prepended to the patch sequence.
- **Transformer Encoder:** The sequence passes through 12 layers with multi-head self-attention and feed-forward networks.
- **MLP Head:** The [CLS] output is classified using a dense layer and softmax.

We apply the early stopping mechanism both per epoch and per round to help the model avoid overfitting, save computational resources, and improve its generalization capability. We use a patience of 3 epochs. If the local validation accuracy does not improve for 3 consecutive epochs, local training for that round is stopped. We use the minimum delta of 0.0001 to monitor the accuracy of the global model. This setting means that if the accuracy does not improve at least 0.01% for 3 consecutive epochs, that round is stopped. These values indicate a common empirical balance, providing the model adequate time to evade possible plateaus while avoiding overfitting and wasting computational resources

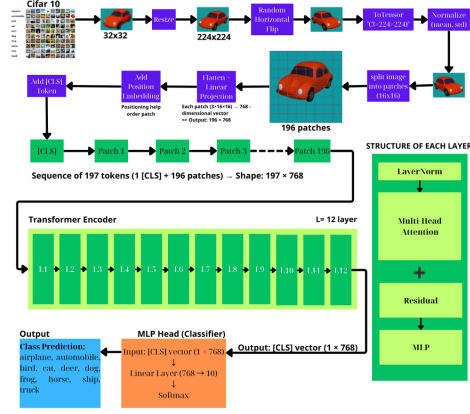


Fig. 2: Vision Transformer (ViT-Tiny) Architecture for CIFAR-10 Classification

on a fully converged or non-converging model. The optimization process employs the Adam optimizer with a learning rate of 3×10^{-4} and weight decay of 10^{-4} , combined with a cosine annealing learning rate scheduler. The loss function is weighted Cross-Entropy, where class weights are computed from the dataset distribution to address class imbalance. The operational workflow of the proposed framework is detailed in Algorithm 1.

B. Privacy Mechanism: Gaussian Differential Privacy

We apply two types of Gaussian noise to enhance privacy and generalization:

- **Input-level noise (data augmentation):** During local training, each client applies a custom Lambda transform that injects zero-mean Gaussian noise into input images as described in Eq. (3):

$$\tilde{x} = x + \mathcal{N}(0, \sigma^2) \quad (3)$$

where x is the original input and $\tilde{x}$ is the perturbed version. This noise, sampled independently per pixel, acts as a regularizer that enhances generalization and reduces overfitting in both IID and non-IID settings.

- **Model-level noise (differential privacy):** After local training, client i perturbs its model parameters before uploading to the server. Instead of defining a new formulation, we implement the mechanism directly as described in Section III-C. Specifically, the noisy update $\tilde{\theta}_i$ is computed according to Eq. (2). Where σ controls the client-side noise scale. This mechanism provides a lightweight and configurable privacy enhancement without requiring ℓ_2 -norm clipping, aligning with recent practical FL studies [17].

Both mechanisms can be independently enabled for flexible evaluation of data augmentation and differential privacy.

Algorithm 1 Federated Averaging (FedAvg) with Early Stopping. The K clients are indexed by k ; B is the local minibatch size, E is the number of local epochs, η is the learning rate.

Server executes:

```

initialize  $w_0$ 
for each round  $t = 1, 2, \dots$  do
   $S_t \leftarrow \{1, 2, \dots, K\}$  {Select all  $K$  clients}
  for each client  $k \in S_t$  in parallel do
     $w_{t+1}^k \leftarrow \text{ClientUpdate}(k, w_t)$ 
  end for
   $w_{t+1} \leftarrow \sum_{k=1}^K \frac{n_k}{n} w_{t+1}^k$  {FedAvg aggregation as in Eq. (1)}
end for

```

ClientUpdate(k, w): // Run on client k

$\mathcal{B} \leftarrow (\text{split } P_k \text{ into batches of size } B)$

for each local epoch i from 1 to E **do**

for each batch $b \in \mathcal{B}$ **do**

$w \leftarrow w - \eta \nabla \ell(w; b)$ // Update using Adam optimizer

end for

if early stopping condition is met **then**

 break // Stop local training

end if

end for

Add Gaussian noise to w {Client-side DP as in Eq. (2)}

return w to server

V. EXPERIMENTS AND ANALYSIS

A. Experimental Setup

To thoroughly evaluate the performance of our privacy-aware Federated Learning framework, we design multiple experimental configurations focusing on architecture (centralized vs. federated), data distribution (IID vs. Non-IID), and the effect of DP.

1) *Dataset description:* CIFAR-10 [18] is a widely used benchmark dataset for image classification, consisting of 60,000 32×32 RGB images evenly distributed across 10 classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. The dataset is divided into 50,000 training images and 10,000 testing images. Its low resolution, class balance, and manageable size make it an ideal choice for benchmarking both centralized and federated learning models, especially in resource-constrained environments where training speed and generalization are key concerns.

2) *Centralized Learning:* Centralized Learning (CL) setup serves as a baseline representing the upper-bound performance. In this setting, a single ViT-Tiny model is trained on the entire CIFAR-10 dataset with full visibility of all training samples and labels. All 10 classes are uniformly represented, and training is conducted in an end-to-end manner with-

out any constraints related to data decentralization or privacy. This configuration provides a reference point for evaluating the impact of FL architecture and differential privacy mechanisms introduced in later experiments.

3) *Federated Learning*: The FL configuration adopts a synchronous client-server architecture implemented using the Flower framework. The system consists of one central server and $N = 3$ federated clients. Each client independently trains a local model on a distinct subset of the CIFAR-10 dataset and transmits its parameter updates to the server, where aggregation is performed via the FedAvg strategy.

All clients employ the same ViT-Tiny architecture, which is pre-trained on ImageNet and fine-tuned on their respective local data. Prior to training, all CIFAR-10 images are resized to 224×224 and normalized using ImageNet statistics to ensure compatibility with the ViT backbone and consistency across clients.

4) *Data Partitioning Strategies*: To simulate varying degrees of data heterogeneity in federated settings, we employ three partitioning schemes:

- **IID** : All 10 classes are randomly and evenly distributed across clients, ensuring that each client receives a balanced dataset representative of the global distribution.
- **Label-Based Split**: The 10 classes are manually divided into non-overlapping subsets, e.g., [4, 3, 3] for 3 clients. Each client receives all samples from its assigned classes, resulting in a hard partition of the label space.
- **80/10/10 Label Distribution**: This scheme first partitions the 10 classes into three label groups according to a [4, 3, 3] split. Each client is assigned one group as its primary group, receiving 80% of the corresponding data samples. The remaining 20% is equally shared among the two other clients (10% each). This creates realistic label dominance while ensuring partial label overlap across clients.

Among these, the 80/10/10 configuration provides a controlled yet realistic approximation of Non-IID settings, making it ideal for evaluating convergence behavior and robustness under heterogeneous data distributions.

5) *Simulation Environment*: All experiments were conducted on Google Colab — a cloud-based Jupyter notebook environment — with the following configuration: Intel Xeon CPU @ 2.2 GHz (virtualized), NVIDIA Tesla T4 GPU, 12.72 GB of memory, and 68 GB of storage. The experimental environment includes Python and essential libraries such as Python 3, PyTorch 2.6.0, NumPy 1.24, Matplotlib 3.7, Scikit-learn 1.3, and the FL orchestration framework Flower 1.18.0.

6) *Comparative Scenarios*: To comprehensively assess the proposed framework, we organize the experiments into two main stages: without Differential Privacy (DP) and with DP enabled.

(a) *Baseline Comparisons (Without DP)*:

This stage focuses on evaluating the comparative performance of the proposed federated framework against the centralized baseline to determine the efficiency of the decentralized architecture. We specifically aim to quantify the impact of data heterogeneity on model convergence and stability by contrasting the standard IID setting with the label-skewed non-IID (80/10/10) distribution.

To ensure a fair comparison, the Centralized Learning model is trained for 50 epochs. Accordingly, each Federated Learning configuration runs for 50 communication rounds with one local epoch per round. Differential privacy mechanisms are disabled in this phase to isolate the effects of the network architecture and data distribution strategies.

(b) *Privacy-Aware Comparisons (With DP)*: To evaluate the privacy-utility trade-off, we analyze performance with DP enabled. This includes:

- **Input-level noise**: Zero-mean Gaussian noise ($\sigma = 0.1$) is injected into training images via a custom Lambda transform.
- **Model-level noise**: This is the primary DP mechanism. We evaluate the system across a spectrum of noise levels. Based on preliminary experiments, we selected four distinct noise scales: 2.5%, 5%, 7.5%, and 10%. This range was chosen to demonstrate the full spectrum of the trade-off:
 - Low Noise (2.5%): To quantify performance with minimal privacy guarantees.
 - Moderate Noise (5% and 7.5%): To observe the gradual decrease as the privacy is strengthened.
 - High Noise (10%): To identify a probable breaking point at which model utility substantially diminishes, it is essential to establish a practical upper bound for noise within this architecture.
- Training is performed with 3 clients and 1 server, across 25 communication rounds and 10 local epochs per client.
- **FL (IID) vs. FL (IID + DP)**: Evaluate the trade-off between utility and privacy under balanced data.
- **FL (Non-IID) vs. FL (Non-IID + DP)**: Assess the compounded effect of data heterogeneity and DP noise.

B. System Metrics and Evaluation

To enable detailed analysis, we log both model training metrics and training time. The four model

training metrics including Accuracy, Precision, Recall, and F1-score are described in Table I. All metrics are exported to CSV files for reproducibility and visualized using matplotlib and seaborn.

TABLE I: Metric Performance (Weighted Averaging)

Metric	Formula	Definition
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Correctly classified samples over total.
Precision (weighted)	$\sum_i w_i \cdot \frac{TP_i}{TP_i+FP_i}$	Average precision weighted by class support.
Recall (weighted)	$\sum_i w_i \cdot \frac{TP_i}{TP_i+FN_i}$	Average recall weighted by class support.
F1-score (weighted)	$\sum_i w_i \cdot \frac{2PR}{P+R}$	Weighted harmonic mean of precision and recall.

C. Experimental Results

1) *Compare Centralized Learning, Federated Learning with IID, and Federated Learning with non-IID:* We first evaluate model performance across three learning configurations: CL, FL-IID, and FL-NonIID. This comparison reveals how decentralization and data distribution affect model accuracy and training time using the ViT-Tiny backbone.

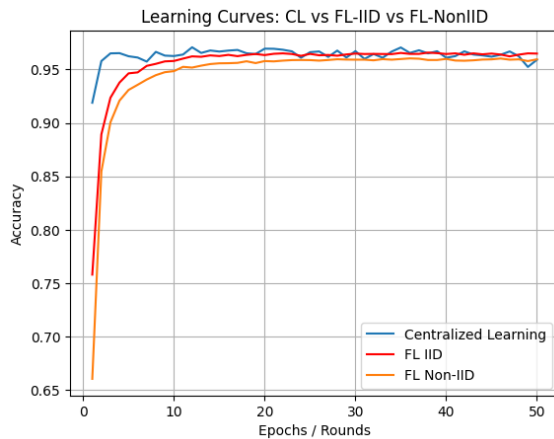


Fig. 3: Accuracy over training epochs across learning settings

As shown in Figure 3, CL acts as the performance upper bound, maintaining high stability (over 96%) while starting at around 92%. FL-IID shows rapid convergence; despite a lower beginning point (over 75%), its figure nearly matches the performance of CL settings within the first 10 rounds, which can be explained by benefiting from the balanced data distribution. FL-NonIID initially falls behind due to data heterogeneity, starting at around 65%. However, it exhibits a robust recovery trend, gradually narrowing the performance gap and achieving competitive accuracy (over 95%) in the subsequent rounds. This

demonstrates that even with limited local training, the proposed framework effectively handles label-skewed distributions.

TABLE II: Performance Comparison across Learning Settings

Metric	CL	FL (IID)	FL (Non-IID)
Accuracy (%)	95.91	96.41	95.91
Precision (%)	95.9	96.41	95.91
Recall (%)	95.91	96.41	95.91
F1-score (%)	95.89	96.4	95.9
Training Time (sec)	12868.38	8967.28	8706.45

Table II summarizes the comparative performance across CL and FL settings. Our experiments show that FL (IID) achieved the highest accuracy of 96.41%, slightly outperforming the CL baseline (95.91%). This suggests that the federated averaging procedure may offer beneficial regularization effects that improve generalization under balanced data distributions. Furthermore, despite the data heterogeneity, the FL (Non-IID) configuration demonstrated remarkable robustness, matching the CL accuracy at 95.91%.

In terms of computational efficiency, the FL settings proved to be significantly faster. While CL required 12,868.38 seconds to complete, FL (IID) and FL (Non-IID) reduce the training time to 8,967.28 seconds and 8,706.45 seconds, respectively. This result represents a time reduction of approximately 30%, demonstrating that the proposed federated framework with ViT-Tiny is computationally efficient when compared to conventional centralized training.

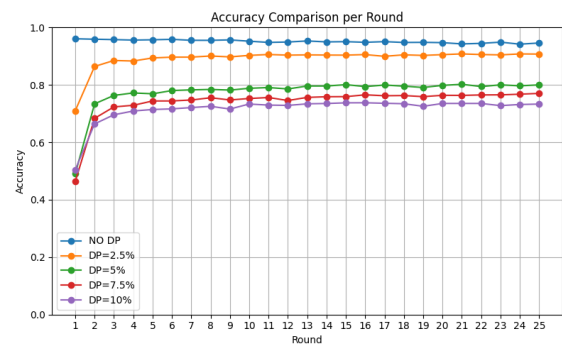


Fig. 4: Accuracy across rounds under different DP noise levels (FL-IID).

2) *Comparison of Federated Learning IID between without DP and with DP:* As illustrated in Figure 4, the No DP setting achieves the highest accuracy (~95%). A mild noise level (2.5%) maintains competitive accuracy (~91%) while higher noise levels (e.g., 5% and above) result in significant accuracy degradation, with the 10% configuration dropping to approximately 72%. This confirms the expected privacy-utility trade-off: moderate noise

preserves model performance, while excessive noise compromises learning efficacy.

TABLE III: Performance and runtime under Gaussian DP (FL-IID)

Setting	Acc (%)	Prec (%)	Rec (%)	F1 (%)	Time (s)
No DP	94.61	94.60	94.61	94.60	41384
DP 2.5%	90.75	90.71	90.75	90.71	38270
DP 5%	80.01	79.97	80.01	79.90	34959
DP 7.5%	77.04	76.91	77.04	76.93	38709
DP 10%	74.34	74.16	74.34	74.20	41096

Table III quantifies the degradation in model performance as the DP noise level increases. Notably, the 2.5% setting provides the best trade-off—maintaining strong accuracy while ensuring privacy. In contrast, noise levels $\geq 7.5\%$ yield significant accuracy losses with only marginal gains in training efficiency.

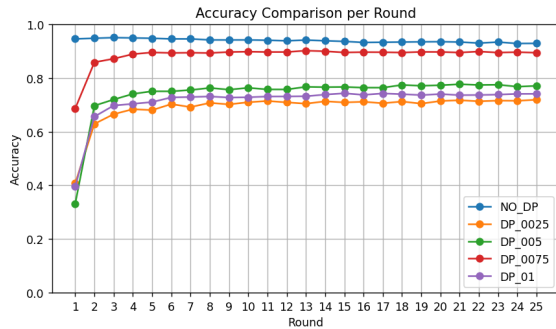


Fig. 5: Accuracy across rounds under different DP noise levels (Non-IID setting)

3) *Comparison of Federated Learning Non-IID between without Differential Privacy and with Differential Privacy:* As depicted in Figure 5, the model trained without DP achieves the highest accuracy (94%), indicating stable convergence in the absence of noise. The 2.5% DP configuration follows closely, maintaining strong utility. However, higher noise levels ($\geq 5\%$) cause significant degradation in accuracy, with the 10% configuration dropping below 71%. This demonstrates the sensitivity of FL models under heterogeneous data to excessive noise injection.

TABLE IV: Performance and training time under DP levels (FL-NonIID)

Setting	Acc (%)	Prec (%)	Rec (%)	F1 (%)	Time (s)
No DP	92.82	92.84	92.82	92.82	40871
DP 2.5%	89.34	89.27	89.34	89.26	39254
DP 5%	77.01	77.16	77.01	76.93	32776
DP 7.5%	74.05	74.05	74.05	73.92	37338
DP 10%	71.86	72.13	71.86	71.83	36536

The results in Table IV confirm this trend: increasing the noise level consistently reduces all perfor-

mance metrics. Among the DP-enabled configurations, 2.5% noise achieves the best balance between privacy and model utility. Interestingly, the 5% noise level results in the shortest training time, while the 10% setting requires longer runtime, possibly due to instability in optimization introduced by excessive noise.

VI. CONCLUSION AND FUTURE WORK

The research addressed three primary contributions, beginning with the proposal of a lightweight, privacy-aware Federated Learning (FL) framework. This framework was built upon the Flower platform and utilized the ViT-Tiny architecture, integrating a configurable, client-side Gaussian-mechanism-based Differential Privacy (DP) mechanism to protect data by injecting calibrated noise into local model parameters before aggregation. Subsequently, we implemented flexible data partitioning strategies, including IID and label-skewed Non-IID (80/10/10 splits), to effectively simulate realistic data heterogeneity scenarios at the network edge. Finally, our comprehensive evaluation along the dimensions of decentralization, data heterogeneity, and privacy demonstrated the system's efficiency and robustness: the FL (IID) configuration achieved the highest accuracy of 96.41%, surprisingly outperforming the Centralized Learning (CL) baseline (95.91%). Furthermore, the federated approach dramatically reduced the training time by approximately 30% when compared to conventional centralized training. The study also determined that a low DP noise scale (e.g., 2.5%) can maintain strong model utility with minimum deterioration.

In the future, we plan to extend this research along several key directions. First, we will incorporate formal privacy accounting using techniques such as Rényi DP and the moments accountant to track the (ϵ, δ) -DP budget across multiple FL rounds. Second, we will scale up the backbone from ViT-Tiny to larger models and evaluate on more challenging datasets like CIFAR-100 or Tiny-ImageNet to better assess scalability. Third, we will integrate more advanced DP mechanisms, including norm clipping, layer-wise or coordinate-wise noise, and adaptive noise schedules to improve convergence. Fourth, we will investigate robustness against Byzantine or poisoning attacks and compare the role of DP with traditional robust aggregation strategies. Finally, we aim to optimize the system for real-world deployment—including intelligent client selection, resource-aware scheduling, and integration into IoT environments—to enable secure and context-aware federated learning at the network edge.

ACKNOWLEDGMENT

This research is funded by University of Information Technology-Vietnam National University

HoChiMinh City under grant number D1-2025-70.

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